Portable IoT-Based Sleep Apnea Monitoring System Using ESP32 for Home Healthcare

ABSTRACT

This project presents a Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare. Sleep apnea is a serious disorder where breathing repeatedly stops and starts during sleep. The system uses sensors such as an IR sensor and heart rate/SpO_2 sensor to continuously monitor the patient's vital parameters. The collected data is processed using an ESP32 microcontroller and transmitted to the ThingSpeak cloud platform for real-time monitoring.The system provides alerts using a vibration motor when abnormal conditions are detected. It also displays real-time values on an LCD screen. This solution enables early detection, continuous monitoring, and improved healthcare support at home.

OBJECTIVE

The main objective of this project is to develop a portable and cost-effective IoT-based system for detecting sleep apnea in real time.The system aims to monitor heart rate, SpO_2 levels, and breathing patterns, provide instant alerts during abnormal conditions, and store data in the cloud for analysis.It also focuses on enabling remote monitoring and improving patient safety in home healthcare environments.

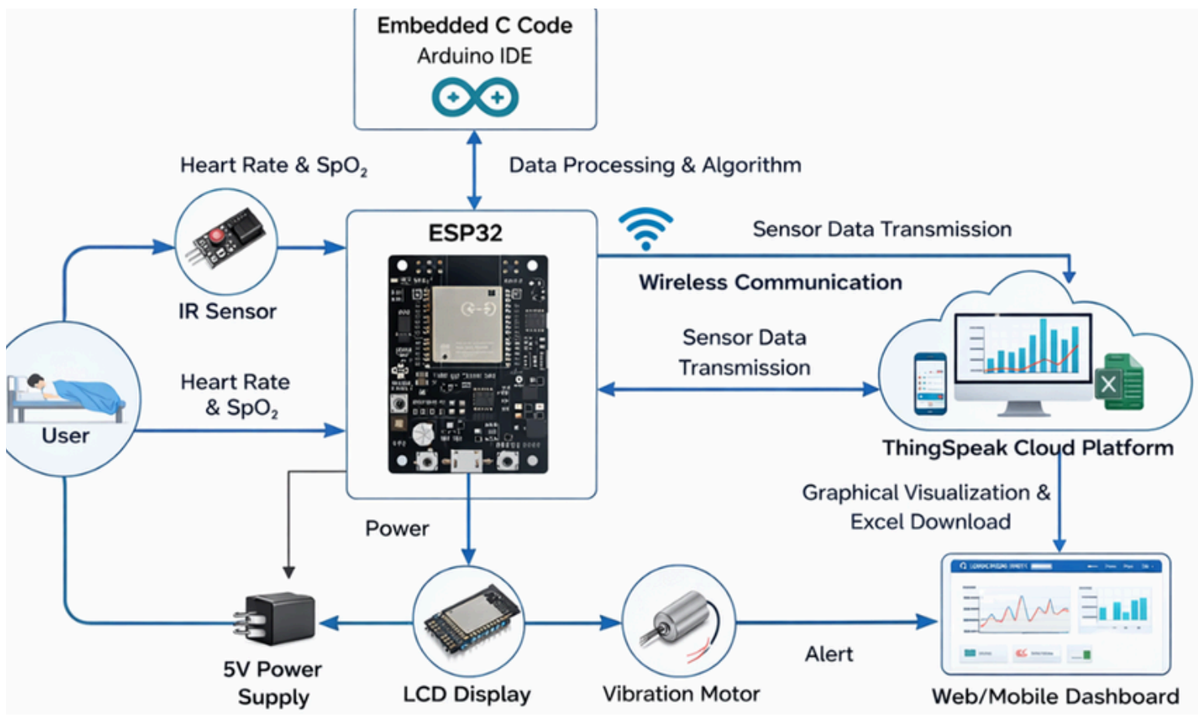
METHODOLOGY

The system works by collecting physiological data using sensors such as IR sensor and pulse oximeter. The ESP32 microcontroller processes the sensor data and analyzes it to detect abnormal conditions related to sleep apnea.The processed data is displayed on an LCD screen and simultaneously sent to the ThingSpeak cloud platform using WiFi. If any abnormality is detected, a vibration motor is activated to alert the user.
The cloud platform stores the data and generates real-time graphs for monitoring and analysis.

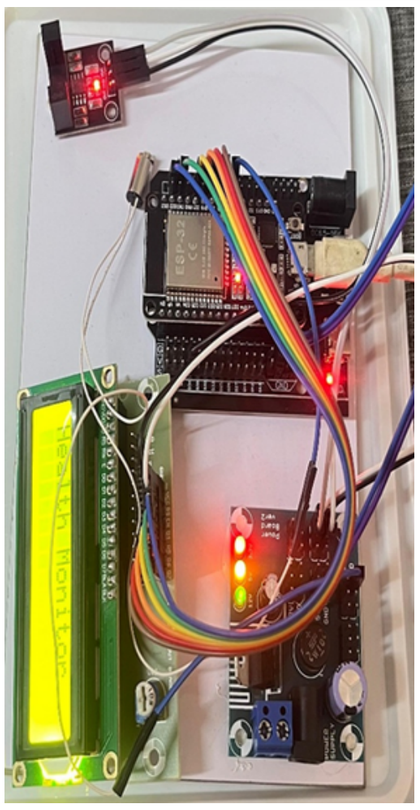
ANALYSIS

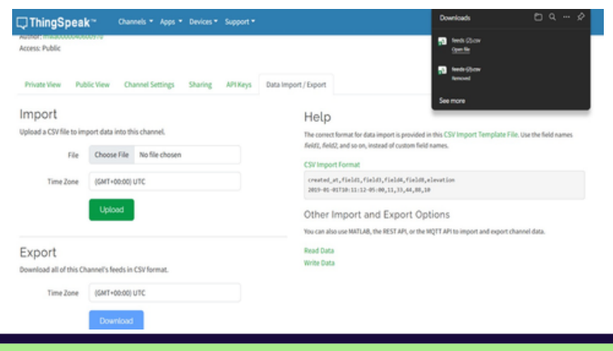
The system improves early detection of sleep apnea by continuously monitoring vital parameters. It reduces the need for hospital-based monitoring and enables home-based healthcare.The use of IoT technology allows real-time data tracking and remote access. It enhances patient safety by providing immediatealertsduringabnormalconditions.The systemiscosteffective,portable,andsuitableforc ontinuousmonitoringapplications.

ARCHITECTURE DIAGRAM



OUTPUT





TECHNOLOGY STACK

- ESP32 Microcontroller
- IR Sensor
- Pulse Sensor (Heart Rate & SpO_2)
- LCD Display
- ThingSpeak Cloud Platform
- Arduino IDEEmbedded C Programming
- WiFi Communication

CONCLUSION

The proposed system provides an efficient and reliable solution for monitoring sleep apnea in home environments. It combines IoT technology, real-time data monitoring, and alert mechanisms to improve patient care.The system helps in early detection of abnormal conditions, reduces health risks, and supports continuous monitoring without hospital dependency. It is a low-cost, portable, and user-friendly solution for modern healthcare needs.

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